United States Patent Application Publication	20250284459
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	CHIOU; Lih-Yih et al.

COMPUTING-IN-MEMORY DEVICE FOR INFERENCE AND LEARNING

Abstract

This disclosure presents a computing-in-memory (CIM) device, including two CIM blocks. These two blocks are connected to the same read bit lines and share an analog-to-digital converter array. The two CIM blocks can perform writing operations simultaneously in one mode. In this case, one block stores a weight matrix, while the other stores the transpose of the weight matrix. In another mode, one CIM block performs the write operation while the other conducts a multiply-accumulate operation. This CIM device can be applied in both inference and training phases.

Inventors: **CHIOU; Lih-Yih (Tainan City, TW), LI; Zong-Han (Kaohsiung City, TW), SHIH; Hong-Ming (Tainan City, TW), HSU; Shun-Hsiu (Taitung County, TW), HUNG; Yi-Hsuan (Nantou County, TW)**

Applicant: **NATIONAL CHENG KUNG UNIVERSITY** (Tainan City, TW)

Family ID: **1000008061141**

Appl. No.: **18/776981**

Filed: **July 18, 2024**

Foreign Application Priority Data

TW	113107981	Mar. 05, 2024

Publication Classification

Int. Cl.: **G06F7/544** (20060101)

U.S. Cl.:

CPC **G06F7/5443** (20130101);

Background/Summary

RELATED APPLICATION

[0001] This application claims priority to Taiwan Application Serial Number 113107981 filed Mar. 5, 2024, which is herein incorporated by reference.

BACKGROUND

Field of Invention

[0002] This disclosure relates to a computing-in-memory (CIM) device capable of being applied to inference and learning.

Description of Related Art

[0003] In recent years, Artificial Intelligence (AI) has been widely applied in academia and everyday life, with the field of Machine Learning (ML) attracting significant attention. Among these, Convolutional Neural Network (CNN) has become a widely used neural network model in areas such as image recognition and object detection. With the advent of the Internet of Things (IoT) era, to reduce the data transmission between edge devices and the cloud, various manufacturers have proposed different software or hardware architectures to decrease computational load or unnecessary data transfer.

[0004] The purpose of Computing-In-Memory (CIM) is to overcome the data transmission bottleneck between the processor and memory in traditional computer systems. In conventional architectures, data storage and processing are separate; the processor needs to retrieve data from memory, process it, and then store the data back in memory. This data round-trip not only consumes time but also increases energy consumption. In contrast, CIM integrates data processing capabilities directly into memory, thereby reducing the need for data transmission and improving efficiency and speed.

SUMMARY

[0005] Embodiments of the present disclosure provide a computing-in-memory (CIM) device including following components. A first memory block includes multiple first CIM units, in which each of the first CIM units is connected to at least one of multiple read bit lines. A second memory block includes multiple second CIM units, in which each of the second CIM units is connected to at least one of the read bit lines. A controlling circuit is configured to write multiple weights of a weight matrix into one of the first memory block and the second memory block, and control the one of the first memory block and the second memory block to perform a multiply and accumulation (MAC) operation. When the first memory block performs the MAC operation, a result of the MAC operation is applied to the read bit lines. When the second memory block performs the MAC operation, the result of the MAC operation is also applied to the read bit lines. An analog-to-digital (ADC) array is connected to the read bit lines and configured to convert multiple analog signals on the read bit lines to multiple digital signals.

[0006] In some embodiments, a number of the weight matrix is greater than 1. The weight matrices correspond to multiple filters respectively. Each of the weights includes multiple weight bits. Each of the first CIM units includes multiple first computing units. Each of the first computing units is connected to one of the read bit lines. The first computing units are arranged as multiple first memory columns and multiple first memory rows. Each of the second CIM units includes multiple second computing units. Each of the second computing units is connected to one of the read bit lines, and the second computing units are arranged as multiple second memory columns and multiple second memory rows.

[0007] In some embodiments, in a simultaneous writing mode, the controlling circuit is configured to write the weights into the first memory block and the second memory block. In the first memory block, the weights of one of the filters with different positions are stored in the first memory rows

respectively, and the weight bits are stored in the first memory columns respectively. In the second memory block, the weights of different ones of the filters with a same position are stored in the second memory rows respectively, and the weight bits are stored in the second memory columns respectively.
[0008] In some embodiments, in a simultaneously writing and calculating mode, the controlling circuit is configured to control one of the first memory block and the second memory block to perform the MAC operation, simultaneously perform a writing procedure on other of the first memory block and the second memory block for writing a portion of the weights.
[0009] In some embodiments, the controlling circuit is configured to control the first memory block and the second memory block to alternatively perform the MAC operation and the writing procedure.
[0010] In some embodiments, the controlling circuit is configured to set an inference phase and a training phase. In the inference phase, the controlling circuit operates in the simultaneously writing and calculating mode.
[0011] In some embodiments, the training phase includes a forward propagation period and a backward propagation period. In the forward propagation period, the controlling circuit operates in the simultaneous writing mode and controls the first memory block to perform the MAC operation.
[0012] In some embodiments, in the backward propagation period, the controlling circuit is configured to control the second memory block to perform the MAC operation and perform the writing procedure on the first memory block or set the first memory block to be in an idle state.
[0013] In some embodiments, the first memory block and the second memory block includes multiple word lines. In the inference phase and the forward propagation, the controlling circuit is configured to apply multiple input features to the word lines. In the backward propagation period, the controlling circuit is configured to apply multiple partial derivatives of a loss with respect to the weights to the word lines.
[0014] In some embodiments, the CIM device further includes an output combiner connected to the ADC array. The output combiner includes multiple adders and a subtractor. The subtractor is configured to receive a most significant bit, and the adders are configured to receive other bits.
[0015] From another aspect, embodiments of the present disclosure provide an electronic device including the aforementioned CIM device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0016] The disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows.
[0017] FIG. **1** illustrates a block diagram of a CIM device according to one embodiment.
[0018] FIG. **2** illustrates a schematic of the MAC operation during forward propagation according to one embodiment.
[0019] FIG. **3** illustrates a schematic of the MAC operation during backward propagation according to one embodiment.
[0020] FIG. **4** illustrates the directional computation of the weights during forward propagation and backward propagation according to one embodiment.
[0021] FIG. **5** illustrates a memory configuration schematic under the simultaneous writing mode according to an embodiment.
[0022] FIG. **6** illustrates a schematic of two memory blocks performing in simultaneously writing and calculating mode according to an embodiment.
[0023] FIG. **7** illustrates the operation of the two memory blocks during the training phase according to an embodiment.

[0024] FIG. **8** illustrates a circuit schematic within a memory block according to an embodiment.
[0025] FIG. **9** illustrates the circuit schematic of an output combiner according to an embodiment.
[0026] FIG. **10** illustrates a schematic of an electronic device according to an embodiment.

DETAILED DESCRIPTION

[0027] Specific embodiments of the present invention are further described in detail below with reference to the accompanying drawings, however, the embodiments described are not intended to limit the present invention and it is not intended for the description of operation to limit the order of implementation. Moreover, any device with equivalent functions that is produced from a structure formed by a recombination of elements shall fall within the scope of the present invention. Additionally, the drawings are only illustrative and are not drawn to actual size.

[0028] The using of "first", "second", "third", etc. in the specification should be understood for identifying units or data described by the same terminology, but are not referred to particular order or sequence.

[0029] FIG. **1** illustrates a block diagram of a CIM device according to one embodiment. Referring to FIG. **1**, a CIM device **100** can be implemented as a chip or a module in a circuit, and can be placed in any suitable electronic device. The CIM device **100** includes a digital time converter (DTC) **110**, an input selector **120**, a first memory block **130**, a second memory block **140**, a weight selector **150**, a writing controller **160**, a CIM controller **170**, an ADC array **180**, and an output combiner **190**. The CIM controller **170** controls the DTC **110**, input selector **120**, weight selector **150**, writing controller **160**, and the ADC array **180**. The input selector **120** connects to the first memory block **130** and second memory block **140** through multiple word lines **122**. Furthermore, the first memory block **130** and second memory block **140** are connected to the ADC array **180** through multiple read bit lines **132**. Notably, the word lines **122** and the read bit line **132** pass through the first memory block **130** and the second memory block **140**, and the relevant setup will be detailed below.

[0030] In other embodiments, multiple circuits in FIG. **1** can be merged together, for example, the DTC **110** and the input selector **120** can be combined into one module, and the weight selector **150** and the writing controller **160** can be merged into one module. In some embodiments, the DTC **110**, input selector **120**, weight selector **150**, writing controller **160**, and CIM controller **170** are collectively referred to as a controlling circuit. Unless specifically mentioned, all data transmission and processing are performed by the controlling circuit, which will not be elaborated further.

[0031] The CIM device **100** is applied in Convolutional Neural Networks. The CIM controller **170** writes multiple weights from at least one weight matrix into the first memory block **130** and the second memory block **140** through the weight selector **150** and the writing controller **160**. The CIM controller **170** also provides input data to the first memory block **130** and the second memory block **140** through the DTC **110** and the input selector **120**. During forward propagation, the input data refer to input features; during backward propagation, the input data are partial derivatives of a loss with respect to the weights. The first memory block **130** and the second memory block **140** perform MAC (Multiply-Accumulate) operations based on the inputs and weights, notably sharing the read bit lines **132**. That is, when the first memory block **130** performs the MAC operation, the result is applied to the read bit lines **132**; similarly, when the second memory block **140** performs the MAC operation, its result is also applied to the read bit lines **132**. Each read bit line **132** transmits a bit, and these operation results are sent in analog form to the ADC array **180**, which converts the analog signals on the read bit lines **132** into digital signals, and then these digital signals are combined by the output combiner **190**.

[0032] The operations of a Convolutional Neural Network include forward propagation and backward propagation. FIG. **2** illustrates a schematic of the MAC operation during forward propagation according to one embodiment. Referring to FIG. **2**, in this embodiment, there are nine filters **201-209**, each containing nine weights represented as w(i,j), where i and j are positive integers, i denotes the filter's sequence number, and j denotes the weight's sequence number. For example, the filter **201** contains nine weights w(0,0) to w(0,8), and so forth. FIG. **2** also shows a part of a feature map **210**. The filter would slide over the feature map **210**. When the filter is at a position **211**, corresponding input features are IF.sub.0 to IF.sub.8, and the related MAC operation is shown in the following Equation 1, where Z.sub.0 is the first result of the filter **201**.

[00001] $Z_0 = \mathrm{IF}_0 \times w(0,0) + \mathrm{IF}_1 \times w(0,1) +$ .Math. $+ \mathrm{IF}_8 \times w(0,8)$ [Equation1]

[0033] On the other hand, during the backward propagation in a training phase, the partial derivatives of the loss with respect to the weights are calculated, which indicate how the weights should be adjusted to reduce the network's prediction error. Based on the chain rule, the calculation must start from the last layer (output layer) of the network, proceed in reverse through each layer, and use the chain rule to compute the partial derivatives for each layer. In each layer, the partial derivatives also undergo MAC operations with the weights. The MAC operation during backward propagation is shown in FIG. **3** which illustrates the partial derivatives d.sub.0˜d.sub.8, while the filters **201-209** remain unchanged. The first result corresponding to the filter **201**, D.sub.0, is calculated as shown in the following Equation 2.

[00002] $D_0 = d_0 \times w(0,0) + d_1 \times w(1,0) +$ .Math. $+ d_8 \times w(8,0)$ [Equation2]

[0034] Comparing FIGS. **2** and **3**, it is evident that both perform MAC operations, but the access sequence of the weights is different. In FIG. **2**, the weights within the same filter are accessed, while in FIG. **3**, the weights at the same position across different filters are accessed. The two MAC operations require the same weights, but the direction of computation is different. Specifically, refer to FIG. **4**, which illustrates the directional computation of the weights during forward propagation and backward propagation according to one embodiment. In FIG. **4**, M represents the number of weights in a filter, and N represents the number of filters, with a matrix **400** storing N×M weights. For example, the first row is arranged with weights w(0,0) to w(0,M), and the first column is arranged with weights w(0,0) to w(N,0), and so on.

[0035] In FIG. **4**, the calculation direction for forward propagation is vertical, while for backward propagation, it is horizontal. Taking the first row as an example, during forward propagation, the input features IF.sub.0 to IF.sub.M are multiplied by weights w(0,0) to w(0,M), and then the results are accumulated through electric currents or charges to yield the result Z.sub.0, while Z.sub.1 is calculated in the second row, and so on. Taking the first column as an example, during backward propagation, the partial derivatives d.sub.0 to d.sub.N are multiplied by weights w(0,0) to w(N,0), and then the results are accumulated through electric currents or charges to yield the result D.sub.0, while D.sub.1 is calculated in the second column, and so on. Due to the different directions of computation, two ADC arrays are required in the known technology because the two directions cannot operate simultaneously, when one ADC array is operational, the other must be idle, leading to a waste of resources.

[0036] In some embodiments, Static Random Access Memory (SRAM) is used for Computing-In-Memory (CIM), where a computing unit (i.e. cell) can perform a calculation for one bit. If a weight consists of 8 bits (referred to as weight bits), then the first row in the matrix **400** can be expanded to be represented as a matrix **410**, where w(0,0)[7] represents the 8th bit of the weight w(0,0), w(0,0)[0] represents the first bit of the weight w(0,0), and so on. Similarly, the second row in the matrix **400** can be expanded to be represented as a matrix **420**.

[0037] In this embodiment, there are two memory blocks, allowing the CIM controller **170** to operate in various modes. These operating modes include a simultaneous writing mode, and a simultaneously writing and calculating mode. Additionally, this setup can be applied in both the inference phase and the training phase. The details of these modes will be further elaborated below.

[0038] In the following embodiment, each input feature and weight has 8 bits. The 8-bit input features are split into 2-bit segments and input into the memory block over four clock cycles, resulting in a 13-bit output for each MAC operation. However, this disclosure does not limit the number of bits for each input feature and weight, nor is it restricted to the aforementioned bitsplitting architecture.

[Simultaneous Writing Mode]

[0039] FIG. **5** illustrates a memory configuration schematic under the simultaneous writing mode according to an embodiment. Referring to FIG. **5**, multiple filters **201-209** are shown, each with a weight matrix. In this embodiment, the size of the weight matrix is 3×3, although this disclosure is not limited to this size. In this embodiment, the first memory block **130** stores the weight matrix, while the second memory block **140** stores the transpose of the weight matrix. The weights required for forward propagation are written to the first memory block **130**, and the weights required for backward propagation are written to the second memory block **140**.

[0040] Specifically, the first memory block **130** comprises multiple CIM units **501-509**. Each of the CIM units **501-509** stores the weights from all positions within a weight matrix. Each CIM unit **501-509** includes multiple computing units **521**, with each computing unit **521** storing a single weight bit. These computing units **521** are arranged in multiple rows (referred to as memory rows) and columns (referred to as memory columns). Taking CIM unit **501** as an example, the first memory column stores the weight bits w(0,0)[7]-w(0,8)[7]. In the single memory column, it stores weight bits of the same sequence order (e.g., all storing the 8th weight bit). The first memory row stores the weight bits w(0,0)[7]-w(0,0)[0]. In the single memory row, it stores bits of the same weight. From another perspective, weights at different positions within filter **201** are stored in various memory rows, while the weight bits are stored in these memory columns. Each computing unit **521** is connected to one read bit line **132**, and the computing units **521** in the same memory column are connected to the same read bit line **132**. In this embodiment, a weight has 8 weight bits, thus CIM unit **501** is connected to 8 read bit lines **132**.

[0041] Similarly, the second memory block **140** contains multiple CIM units **511-519**, each comprising numerous computing units **522** arranged in rows (referred to as memory rows) and columns (referred to as memory columns). Likewise, each computing unit **522** is connected to one read bit line **132**, so the CIM unit **511** is connected to 8 read bit lines **132**. Taking CIM unit **511** as an example, the first memory column stores the weight bits w(0,0)[7], w(1,0)[7] . . . w(8,0)[7]; the first memory row stores the weight bits w(0,0)[7], w(0,0)[6] . . . w(0,0)[0]. This means that the weights at the same position across different filters are stored in the memory rows, while the weight bits of different orders are stored in the memory columns.

[0042] In the example of FIG. **5**, input features IF.sub.0-IF.sub.8 are sent to the first memory block **130**, and partial derivatives d.sub.0 to de are sent to the second memory block **140**. This computation aligns with FIG. **4**, but the operation results are vertically transmitted on the read bit lines **132**. Since the read bit lines **132** are shared, only one memory block can perform the MAC operation at a time. Regardless of which memory block performs the MAC operation, the operation results can be output through the read bit lines **132**.

[0043] The ADC array **180** includes multiple Analog-to-Digital Converters (ADCs) ADC[7]-ADC[0], each of which converts the analog signal on one read bit line **132** into a digital signal. Then, the digital signals corresponding to the CIM units **501-509** are combined together through the output combiner **190**, thereby outputting the partial derivatives D.sub.0-D.sub.8 or the results Z.sub.0-Z.sub.8.

[Simultaneously Writing and Calculating Mode]

[0044] FIG. **6** illustrates a schematic of two memory blocks performing in simultaneously writing and calculating mode according to an embodiment. Referring to FIG. **6**, this mode requires only forward propagation or backward propagation. In some embodiments, the number of weights is too large to load all at once, necessitating the loading of a subset of weights for the MAC operation, followed by the next batch of weights. Having two memory blocks allows for simultaneous writing and calculating, effectively reducing the overall computation time by hiding the writing time within the computing time. At any given time interval, at most one memory block can perform the MAC operation. For example, multiple time intervals **601-605** can be set. During the time interval **601**, the first memory block **130** performs a writing procedure to load a subset of weights, while the second memory block **140** remains idle. During the time interval **602**, the first memory block **130** performs the MAC operation, while the second memory block **140** carries out the writing procedure. During the time interval **603**, the first memory block **130** undergoes the writing procedure, while the second memory block **140** performs the MAC operation, and so on. Notably, when the first and second memory blocks **130** and **140** perform forward propagation together, both store the weight matrix in the same orientation; if they perform backward propagation together, both store the transpose of the weight matrix. Whether for forward or backward propagation, the first and second memory blocks **130** and **140** alternatively perform the MAC operation and the writing procedure. This architecture fully utilizes the ADC array **180**, enhancing overall efficiency and reducing computation latency. Compared to the known technology that uses two sets of DTC and ADC arrays, the embodiment in FIG. **6** shares these hardware components, which can reduce circuit area and power consumption, achieving up to a 28% reduction in power consumption in some experiments.

[Overall Operations]

[0045] Convolutional Neural Network operations can be divided into three scenarios. The first scenario is the inference phase, where the controlling circuit operates in simultaneously writing and calculating mode. In this scenario, weights are alternately loaded into the first memory block **130** and the second memory block **140**, and the MAC operations are performed based on these weights and the input features. The operational schematic is illustrated in FIG. **6**.

[0046] The second scenario is the forward propagation during the training phase, and the third scenario is the backward propagation during the training phase. Both scenarios can be realized using the two memory blocks. FIG. **7** illustrates the operation of the two memory blocks during the training phase according to an embodiment. Referring to FIG. **7**, during the forward propagation, the controlling circuit operates in simultaneous writing mode. In FIG. **7**, “writing (forward)” refers to writing weights in the order required for forward propagation, while “writing (backward)” refers to writing weights in the order required for backward propagation. During a time interval **701**, both the first memory block **130** and the second memory block **140** perform the writing procedure, where the written weights follow the arrangement shown in FIG. **5**. During a time interval **702**, the first memory block **130** performs the MAC operation, while the second memory block **140** is idle. Time intervals **703-705** can be inferred similarly.

[0047] During the backward propagation, the controlling circuit controls the second memory block **140** to perform the MAC operation, while the first memory block **130** either undergoes a writing procedure or is set to an idle state. Specifically, during time intervals **711-713**, the second memory block **140** performs the MAC operation, and the first memory block **130** undergoes a writing procedure, such as writing updated weights to prepare for the forward propagation of the next sample. During a time interval **714**, the second memory block **140** continues the MAC operation, while the first memory block **130** remains idle. Thus, the CIM device **100** can be used for both the inference phase and the training phase, and the shared use of the ADC array **180** can save circuit area and power consumption.

[0048] FIG. **8** illustrates a circuit schematic within a memory block according to an embodiment. Here, the first memory block **130** is used as an example, but the circuit structure of the second memory block **140** is the same as that of the first memory block **130**, hence it is not elaborated further. The first memory block **130** contains multiple computing units **521** arranged in a matrix. The number of rows in the matrix is equal to the number of bits in a weight multiplied by the number of filters, which is 8×9=72 rows in this embodiment. The number of columns in the matrix corresponds to the number of weights in a filter, which is 9 columns in this embodiment. Each computing unit **521** is connected to a read bit line **132** and a word line **122**, with each read bit line **132** connected to a reset switch Rst. Each computing unit **521** includes a Static Random Access Memory (SRAM) cell **811**, a switch **812**, and a switch **813**. The SRAM cell **811** contains six transistors, while switches **812** and **813** are each a single transistor, thus the computing unit **521** can also be referred to as an 8T (eight-transistor) SRAM computing unit.

[0049] The SRAM cell **811** stores a single weight bit. When this weight bit is “1”, it turns on the switch **812**, and conversely, it turns off the switch **812**. On the other hand, when performing a MAC operation, the switch **813** can be turned on or off based on the signal on the word line **122**. Specifically, during the forward propagation, each bit of the input features is applied to a word line, turning on the switch **813** if this bit is “1”. If the weight bit is also “1”, a current will be generated from the system voltage VDD into the corresponding read bit line **132**, and the currents generated by the computing units **521** on the same column will accumulate. If the bit of the input features is “0” or the weight bit is “0”, no current will flow into the read bit line **132**. During the backward propagation, the partial derivatives of the loss with respect to the weights are applied to the word lines **122**, and the operation is similar to the forward propagation. Using 8T SRAM for the computing unit has the advantage of a larger static noise margin, allowing for operation at lower voltages to save power consumption. Another advantage is that the parasitic capacitance on the read bit line **132** is not a stable value and can change due to manufacturing or layout differences. Therefore, by fine-tuning the operating voltage of the SRAM, it can provide a consistent current to the read bit line **132** under different process variations.

[0050] Referring to FIG. **1**, the analog signals on the read bit lines **132** are transmitted to the ADC array **180**, which includes multiple Analog-to-Digital Converters (ADCs) that convert the analog signals on the read bit lines **132** into digital signals. In this example, a digital signal consists of 4 bits. Any type of Analog-to-Digital Converter can be used in this context, and this disclosure is not limited to a specific type of ADC.

[0051] In this embodiment, MAC operations are performed on a bit-by-bit basis, which offers better resistance to process variations. This is advantageous because the amount of data that needs to be quantized is reduced, and the process can be managed using digital circuitry, avoiding unnecessary errors. However, this setup requires an output combiner **190** to merge the outputs from different weights. FIG. **9** illustrates the circuit schematic of an output combiner according to an embodiment. Referring to FIG. **9**, take a CIM unit as an example, there are eight digital signals **901-908**, each consisting of 4 bits. The output combiner **190** includes multiple shifters **911-918** that shift the digital signals **901-908**, respectively. The output combiner **190** also includes a subtractor **921** and multiple adders **922-927**. The subtractor **921** receives the most significant bit, and the remaining bits are processed by the adders **922-924**, then further processed by the adders **925-927**, resulting in a 12-bit signed output **930**. In this embodiment, the output **930** is signed, indicating it can represent both positive and negative values.

[0052] FIG. **10** illustrates a schematic of an electronic device according to an embodiment. Referring to FIG. **10**, from another perspective, this disclosure also provides an electronic device **1000** that includes the aforementioned CIM device **100**. The electronic device **1000** can be implemented as a smartphone, tablet, laptop, other forms of mobile devices, home appliances, etc., and is not limited to these examples.

[0053] Although the present invention has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein. It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present invention without departing from the scope or spirit of the invention. In view of the foregoing, it is intended that the present invention cover modifications and variations of this invention provided they fall within the scope of the following claims.

Claims

1. A computing-in-memory (CIM) device, comprising: a first memory block, comprising a plurality of first CIM units, wherein each of the first CIM units is connected to at least one of a plurality of read bit lines; a second memory block, comprising a plurality of second CIM units, wherein each of the second CIM units is connected to at least one of the read bit lines; at least one controlling circuit, configured to write a plurality of weights of at least one weight matrix into one of the first memory block and the second memory block, and control the one of the first memory block and the second memory block to perform a multiply and accumulation (MAC) operation, wherein when the first memory block performs the MAC operation, a result of the MAC operation is applied to the read bit lines, wherein when the second memory block performs the MAC operation, the result of the MAC operation is applied to the read bit lines; and an analog-to-digital (ADC) array, connected to the read bit lines and configured to convert a plurality of analog signals on the read bit lines to a plurality of digital signals.
2. The CIM device of claim 1, wherein a number of the at least one weight matrix is greater than 1, the weight matrices correspond to a plurality of filters respectively, each of the weights comprises a plurality of weight bits, each of the first CIM units comprises a plurality of first computing units, each of the first computing units is connected to one of the read bit lines, and the first computing units are arranged as a plurality of first memory columns and a plurality of first memory rows, wherein each of the second CIM units comprises a plurality of second computing units, each of the second computing units is connected to one of the read bit lines, and the second computing units are arranged as a plurality of second memory columns and a plurality of second memory rows.
3. The CIM device of claim 2, wherein in a simultaneous writing mode, the at least one controlling circuit is configured to write the weights into the first memory block and the second memory block, wherein in the first memory block, the weights of one of the filters with different positions are stored in the first memory rows respectively, and the weight bits are stored in the first memory columns respectively, wherein in the second memory block, the weights of different ones of the filters with a same position are stored in the second memory rows respectively, and the weight bits are stored in the second memory columns respectively.
4. The CIM device of claim 3, wherein in a simultaneously writing and calculating mode, the at least one controlling circuit is configured to control one of the first memory block and the second memory block to perform the MAC operation, simultaneously perform a writing procedure on other of the first memory block and the second memory block for writing a portion of the weights.
5. The CIM device of claim 4, wherein the at least one controlling circuit is configured to control the first memory block and the second memory block to alternatively perform the MAC operation and the writing procedure.
6. The CIM device of claim 5, wherein the at least one controlling circuit is configured to set an inference phase and a training phase, wherein in the inference phase, the at least one controlling circuit operates in the simultaneously writing and calculating mode.
7. The CIM device of claim 6, wherein the training phase comprises a forward propagation and a backward propagation, wherein in the forward propagation, the at least one controlling circuit operates in the simultaneous writing mode and control the first memory block to perform the MAC operation.
8. The CIM device of claim 7, wherein in the backward propagation, the at least one controlling circuit is configured to control the second memory block to perform the MAC operation and perform the writing procedure on the first memory block or set the first memory block to be in an idle state.
9. The CIM device of claim 8, wherein the first memory block and the second memory block comprises a plurality of word lines, wherein in the inference phase and the forward propagation, the at least one controlling circuit is configured to apply a plurality of input features on the word lines, wherein in the backward propagation, the at least one controlling circuit is configured to

apply a plurality of partial derivatives of a loss with respect to the weights to the word lines.

10. The CIM device of claim 9, further comprising: an output combiner, connected to the ADC array, wherein the output combiner comprises a plurality of adders and a subtractor, and the subtractor is configured to receive a most significant bit, and the adders are configured to receive other bits.

11. An electronic device, comprising: a computing-in-memory (CIM) device, comprising: a first memory block, comprising a plurality of first CIM units, wherein each of the first CIM units is connected to at least one of a plurality of read bit lines; a second memory block, comprising a plurality of second CIM units, wherein each of the second CIM units is connected to at least one of the read bit lines; at least one controlling circuit, configured to write a plurality of weights of at least one weight matrix into one of the first memory block and the second memory block, and control the one of the first memory block and the second memory block to perform a multiply and accumulation (MAC) operation, wherein when the first memory block performs the MAC operation, a result of the MAC operation is applied to the read bit lines, wherein when the second memory block performs the MAC operation, the result of the MAC operation is applied to the read bit lines; and an analog-to-digital (ADC) array, connected to the read bit lines and configured to convert a plurality of analog signals on the read bit lines to a plurality of digital signals.

12. The electronic device of claim 11, wherein a number of the at least one weight matrix is greater than 1, the weight matrices correspond to a plurality of filters respectively, each of the weights comprises a plurality of weight bits, each of the first CIM units comprises a plurality of first computing units, each of the first computing units is connected to one of the read bit lines, and the first computing units are arranged as a plurality of first memory columns and a plurality of first memory rows, wherein each of the second CIM units comprises a plurality of second computing units, each of the second computing units is connected to one of the read bit lines, and the second computing units are arranged as a plurality of second memory columns and a plurality of second memory rows.

13. The electronic device of claim 12, wherein in a simultaneous writing mode, the at least one controlling circuit is configured to write the weights into the first memory block and the second memory block, wherein in the first memory block, the weights of one of the filters with different positions are stored in the first memory rows respectively, and the weight bits are stored in the first memory columns respectively, wherein in the second memory block, the weights of different ones of the filters with a same position are stored in the second memory rows respectively, and the weight bits are stored in the second memory columns respectively.

14. The electronic device of claim 13, wherein in a simultaneously writing and calculating mode, the at least one controlling circuit is configured to control one of the first memory block and the second memory block to perform the MAC operation, simultaneously perform a writing procedure on other of the first memory block and the second memory block for writing a portion of the weights.

15. The electronic device of claim 14, wherein the at least one controlling circuit is configured to control the first memory block and the second memory block alternatively perform the MAC operation and the writing procedure.